

- 5 (a) Define potential at a point in an electric field.

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.....[2]

- (b) An isolated metal sphere of radius r carries a charge $+Q$. The charge may be assumed to be concentrated at the centre of the sphere.

- (i) State, in terms of r and Q , the electric potential V at the surface of the sphere. Identify any other symbols you use.

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- (ii) Write down the relationship between capacitance C , charge Q and potential V .

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- (iii) Hence show that the capacitance C of the sphere is given by

$$C = 4\pi\epsilon_0 r.$$

- (c) The sphere in (b) has a radius of 15 cm and carries a charge of $2.0 \times 10^{-6} \text{ C}$.

Calculate

- (i) the capacitance of the sphere,

capacitance = μF

- (ii) the energy stored on the sphere.

energy = J
[4]